

- Q.28 Explain the function and working of reverse circuit breakers.
- Q.29 Give various parts of Ni-Cd battery.
- Q.30 Explain parallel operation maintenance of AC generator.
- Q.31 Draw a neat labeled diagram of the landing gear circuit.
- Q.32 Explain the working of starter motors.
- Q.33 Explain the application of high pass filters with example.
- Q.34 Where is the function of thermocouple function. Give example.
- Q.35 Write short note on Auto xmer

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Differentiate w.r.t construction and working of ammeter, voltmeter and wattmeter.
- Q.37 Explain different types of AC motors and their working principles.
- Q.38 Write short notes on
- Transformers
 - Rectifiers.

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5th Sem / AME Subject:- Aircraft Electrical Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is not a cause of static electricity on aircraft.
- Friction with air
 - Fuel transfer
 - Solar radiation
 - Precipitation static
- Q.2 A step-up transformer has a turns ratio (K) such that.
- $K < 1$
 - $K > 1$
 - $K = 0$
 - $K = 1$
- Q.3 The Reason for the rapid wear of brushes is _____
- Abrasion from dust
 - Excessive spring pressure
 - Rough commutator bars
 - Abrasion from dust, excessive spring pressure and rough commutator bars
- Q.4 Vibration galvanometers are used for _____
- very high frequency
 - very low frequency
 - low audio frequency
 - high audio frequency
- Q.5 The minimum number of conductor strands required in aircraft quality wire is _____?

- a) 19 Strands b) One strand
c) 7 Strands d) 14 strands
- Q.6 The armature voltage control of DC motor provides _____
a) Constant power drive
b) Constant voltage drive
c) Constant current drive
d) Constant torque drive
- Q.7 Rectifier is used to correct the:
a) DC voltage level b) AC voltage level
c) AC to DC d) DC to AC
- Q.8 Emf and torque produces in a DC machine are proportional to _____ and _____ respectively.
a) Armature speed and armature emf
b) Armature emf and armature speed
c) Armature current and armature emf
d) Armature speed and armature current
- Q.9 At low frequency, sources can be one of the following
a) amplifiers b) regulators
c) oscillators d) op amps
- Q.10 The number of commutator bars equal to for both lap and wave windings, is equal to
a) no of Slots
b) no of armature conductors
c) Windings elements
d) no of Poles

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 A dynamometer wattmeters uses _____ and _____ coils to measure power.
- Q.12 The corona effect is primarily caused by _____
- Q.13 Efficiency is transformers is maximized when iron losses are equal to _____ losses.
- Q.14 The fastest operating fuse is _____
- Q.15 Give one advantage of electrical energy over other fuels.
- Q.16 Give one difference of AC and DC.
- Q.17 Give the purpose of shielding.
- Q.18 Paralleling of generators enables _____
- Q.19 They are _____ types of DC motors.
- Q.20 A Quartz crystal is used in _____ transducer.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Explain the construction and working of dynamometer type instruments.
- Q.22 Classify the draw various types of terminals.
- Q.23 Explain the construction and function of relays.
- Q.24 Describe P-static cause and prevention.
- Q.25 What are the various types of switches and their application?
- Q.26 Write short note on current limiters.
- Q.27 Give the construction of voltage regulators.